Vikram Kumar

Senior Full Stack Developer

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Professional Profile

Senior **Full Stack Developer** with **12+ years of professional experience** delivering **complex software solutions** in **Agile team environments**. Previous **Government of Alberta** and **Alberta Energy Regulator** experience, plus enterprise delivery at IBM, GSTS, and Inovi Technologies. Production experience with **modern application frameworks** and **frontend JavaScript frameworks**, including **React**, **Angular**, **Node.js**, **Express**, and **NestJS**. Skilled in **collaborative Git source control**, **automated unit testing** and **integration testing**, **relational and NoSQL databases**, **CI/CD**, design patterns, and **Azure**/**AWS** architecture. Hands-on **Generative AI** training and evaluation includes **prompt design** and **critical evaluation of AI outputs** for accuracy, **security risks**, and **hallucination patterns**.

Technical Skills

Frontend: React, **Angular**, **AngularJS**, **TypeScript**, **JavaScript**, **Redux Toolkit**, **NgRx**, **RxJS**; responsive single-page applications and reusable UI components.

Backend and APIs: Node.js, **Express**, **NestJS**, **Python**, **Java Spring Boot**, **C#**, **REST**, **GraphQL**, **Prisma**, **TypeORM**.

Databases and testing: PostgreSQL, **MySQL**, **T-SQL**, **Azure SQL**, **MongoDB**, **Cosmos DB**, **DynamoDB**, **Redis**; **Jest**, **React Testing Library**, **Supertest**, **Cypress**, **JUnit**, **Mockito**.

Cloud and delivery: Azure, **AWS**, **Azure DevOps**, **Git**, **GitHub Actions**, **Bitbucket**, **Docker**, **Kubernetes**, **AKS**, **Terraform**, **CloudFormation**; **CI/CD**, **workflow automation**, **software design patterns**.

Security and operations: Azure AD, **RBAC**, **OAuth2**, **OpenID Connect**, **Key Vault**, **Blob Storage**, **Azure Functions**, **App Service**, **Application Insights**, **Log Analytics**; monitoring, alerting, and distributed systems troubleshooting.

Professional Experience

Alberta Energy Regulator

Senior Software Developer

12 Jan 2026 - 31 Aug 2026 | Calgary, Canada

Azure Data Engineering and Platform Delivery

- Designed and delivered enterprise-grade data ingestion workflows for bulk submissions, covering file intake, schema-aware parsing, transformation, staging, validation, duplicate detection, and loading into **relational SQL databases**.
- Built reusable **Node.js**/**TypeScript** pipeline components that dynamically map spreadsheet headers to database parameters, reducing hardcoded transformation logic and improving data accuracy across submission types.
- Integrated **Azure Blob Storage** for structured file persistence and **Azure Key Vault** for secure runtime secret resolution across development, test, and production environments.
- Implemented automated **Azure DevOps** deployment workflows for containerized services using **Docker**, **Kubernetes** overlays, environment-specific configuration, and **Nginx** reverse-proxy routing.
- Troubleshot and strengthened **CI/CD** reliability by resolving missing deployment overlays, environment configuration gaps, **Key Vault** secret issues, and production routing/session problems.
- Developed operational telemetry and real-time progress patterns using **Server-Sent Events** and production logging to improve observability of long-running ingestion processes.
- Applied production engineering practices including repeatable deployments, environment isolation, secret management, scalable container orchestration, data validation, and operational monitoring.
- Wrote and optimized **T-SQL** stored procedures for staging validation and data loading, including dynamic SQL, type conversion, uniqueness checks, and high-volume data integrity rules.

Frontend and Application Engineering

- Built production **React**/**TypeScript** single-page applications for multi-step bulk data submission, securities estimation, administrative review, and public-facing data workflows.
- Integrated **MSAL** with **Azure AD** authentication and role-based protected routes; resolved redirect loops, token acquisition issues, and session-storage conflicts.
- Built a custom virtualized spreadsheet grid for large datasets, improving **user experience (UX)** through inline validation, duplicate-row detection, error highlighting, and responsive feedback.
- Used **Ant Design** and **Tailwind CSS** to create reusable shared components, responsive filtering experiences, form workflows, and calculation-driven user interfaces.
- Configured **Vite** and **Nginx** for performant SPA builds and containerized production deployments.

Full Stack Delivery

- Delivered full-stack submission workflows spanning **React** user interfaces, reusable **Node.js**/**TypeScript** ingestion components, relational data validation, secure **Azure** services, and containerized production deployment.
- Applied **separation of concerns** across reusable UI components, transformation logic, SQL validation, and environment configuration to keep complex submission workflows maintainable.

Mercor

Senior Software Engineer - AI Model Training

01 May 2025 - 30 Dec 2025 | Remote

- Contributed to **Generative AI** model training and evaluation, assessing **large language models** (**LLMs**) across multiple programming languages and software engineering domains.
- Wrote production-quality code solutions in **Python**, **C#**, **Node.js**, and **Java** to generate high-fidelity training data for **AI models**.
- Authored detailed rubrics defining evaluation criteria for correctness, code quality, performance, security, and reasoning clarity - used to standardize assessments across the team.
- Critically evaluated and ranked **AI-generated code** responses for accuracy, correctness, and consistency, providing structured feedback to improve model performance across iterations.
- Identified and documented subtle bugs, logic errors, edge cases, and anti-patterns in **model-generated code** across a wide range of complexity levels.
- Covered diverse problem domains including **REST** API design, database query optimization, algorithms and data structures, object-oriented design, and system architecture.
- Applied language-specific best practices (e.g. async patterns in **Node.js**, LINQ in **C#**, OOP principles in **Java**, data manipulation in **Python**) to ensure training examples met a high engineering standard.
- Collaborated with prompt engineers and QA reviewers on **prompt design** and refinement, clearer evaluation rubrics, and consistent assessment standards.
- Contributed to iterative model improvement cycles by flagging recurring failure patterns and suggesting targeted training adjustments.
- Evaluated AI outputs for **security risks**, including SQL injection, improper error handling, and insecure API patterns, and flagged vulnerabilities as training feedback.
- Evaluated algorithmic efficiency and time/space complexity in model-generated solutions, ensuring training data reflected optimal and scalable approaches.
- Reviewed model outputs for code readability and maintainability, including naming conventions, **separation of concerns**, modularity, and inline documentation quality.
- Contributed to **prompt engineering** by analyzing model failure patterns and recommending clearer or more constrained task formulations to improve output quality.
- Handled tasks spanning beginner to expert difficulty levels, requiring broad and deep knowledge across backend, full-stack, and systems programming concepts.
- Provided nuanced assessments on ambiguous or open-ended problems where multiple valid solutions existed, exercising strong engineering judgment to rank trade-offs.
- Maintained high throughput and consistency across large volumes of daily evaluations while upholding quality standards set by rubric guidelines.

- Worked in a fully remote, asynchronous environment requiring strong self-management, clear written communication, and disciplined prioritization of tasks.
- Independently designed and wrote rubrics from scratch for a wide range of software engineering tasks, defining clear scoring criteria across dimensions such as correctness, efficiency, security, readability, and completeness.
- Ensured rubrics were unambiguous and consistently applicable across different evaluators, reducing subjectivity in model assessments.
- Tailored rubrics to language-specific and domain-specific contexts (e.g. different criteria for a **Python** data pipeline vs. a **C# REST** API vs. a **Java** OOP design problem).
- Iterated on rubrics based on evaluator feedback and edge cases encountered during scoring, continuously refining criteria to improve assessment reliability.
- Balanced strictness and flexibility in **rubric design** - strict enough to catch model errors consistently, flexible enough to fairly credit alternative valid approaches.
- Contributed rubrics that were adopted by the broader team, serving as reference standards for evaluating similar task categories.
- Analyzed how **LLMs** reason about code, including common failure modes, **hallucination patterns**, and problems that challenge model capabilities.

Government of Alberta

Senior Software Developer

02 Oct 2023 - 15 Apr 2025 | Calgary, Canada

Enterprise Applications and Digital Service Delivery

- Designed and maintained enterprise data-processing and **workflow automation** services for the **Government of Alberta** on **Azure**, using **Azure Functions**, **Logic Apps**, **Cosmos DB**, **Redis**, relational data services, and **TypeScript**-based connectors for transformation and orchestration.
- Applied **software design patterns** and validation practices, including **DTO validation**, **repository patterns**, **schema validation**, and controlled service contracts, to improve data accuracy, integrity, and maintainability across financial and cross-service workflows.
- Built and operated scalable production **microservices** on **Azure Kubernetes Service** (**AKS**) and **Azure App Service** using **Node.js**, **NestJS**, **GraphQL**, **TypeORM**, and **Azure Functions**, with containerized deployment and modular service boundaries.
- Established multi-stage **CI/CD** pipelines and infrastructure automation using **Azure DevOps Pipelines**, YAML workflows, **Terraform**, **GitHub Actions**, quality gates, and environment-specific deployment controls for repeatable releases.
- Implemented production monitoring and alerting using **Azure Monitor**, **Application Insights**, **Log Analytics**, and **KQL** dashboards to track application performance, failures, SLA compliance, and operational health.
- Built asynchronous, event-driven integration pipelines with **Azure Service Bus**, dead-letter queues, message contracts, and **JSON Schema** validation to support reliable enterprise data movement between services.
- Used **Azure API Management** to govern **REST** and **GraphQL** APIs through throttling, traffic policies, key rotation, and secure service exposure across enterprise applications.
- Applied **Kubernetes**, automated deployment, monitoring, data quality controls, and cloud orchestration to support reliable enterprise application delivery on **Azure**.
- Worked across **relational and NoSQL data technologies**, combining relational services and **TypeORM** with **Cosmos DB** and **Redis** in enterprise application and data-processing workflows.
- Designed service integration around explicit API and message contracts, reusable **TypeScript** connectors, and repository-based access to data, supporting maintainable **distributed application architecture**.
- Combined automated deployment controls, secure API governance, data validation, monitoring, and alerting to support reliable **Government of Alberta** production services.

GSTS

Senior Software Developer - Full Stack

23 May 2023 - 08 Aug 2023 | Halifax, Canada | Remote

- Modernized a **legacy monolithic Express backend** into cost-efficient, event-driven digital services using **AWS Lambda**, preserving server-based APIs through embedded **Express** and maintaining existing application integration points.
- Designed hybrid APIs with **Java Spring Boot microservices** and **Node.js** Lambdas, optimizing compute-heavy and real-time workloads.
- Managed secure, scalable data using **AWS DynamoDB** and **Amazon S3**, with **Spring Boot**, **Spring Data**, and **AWS SDKs** for cross-runtime communication.
- Implemented fine-grained access control using **AWS IAM** and **Amazon Cognito**, integrating **OAuth2** with **Spring Security** and Cognito-hosted UIs.
- Built user group logic with **AWS Lambda** and **Cognito User Pools**, mirrored in **Spring Boot REST** endpoints for admin operations.
- Developed GIS-enabled frontends using **Leaflet.js** and **Mapbox**, delivering real-time spatial data via **AWS S3** and **Spring Boot** geospatial APIs.
- Enabled geospatial precision using **AWS Lambda** geometry functions for compliance validation.
- Triggered geofence alerts via **AWS SNS**, processed in **Spring Boot** event consumers.
- Synced real-time vessel data using **AWS AppSync**, **GraphQL**, Lambda, and **Spring Boot GraphQL** gateways.
- Enabled PDF/CSV import/export workflows using **AWS S3**, validated and parsed via **Spring Boot** and **Node.js** hybrid pipelines.
- Implemented **automated unit testing** with **Jest**, **JUnit**, and **Mockito**, and **end-to-end testing** with **Cypress**, to support reliable **JavaScript** and **Spring Boot** applications in **AWS**-native environments.
- Automated **CI/CD** build and deployment pipelines with **AWS CodePipeline**, **Maven** builds for **Java** services, and **GitHub Actions** for frontend releases.
- Achieved scalable, cross-platform architecture by bridging **Spring Boot** APIs and **Node.js** Lambda functions.
- Delivered full-stack maritime application capabilities across GIS-enabled interfaces, **REST** and **GraphQL** APIs, event-driven **Node.js**/**Express** services, **Java Spring Boot microservices**, and **NoSQL** data storage.
- Applied distributed and **event-driven architecture** patterns to vessel updates, geospatial validation, and geofence alerts across **AWS** and **Spring Boot** services.

IBM

Application Developer - Full Stack

04 Apr 2022 - 12 Apr 2023 | Calgary, Canada

Lead Developer for Air Canada

- Mentored developers in **React**, npm, and **TypeScript**, and **modernized legacy code** to improve application performance, readability, and maintainability.
- Architected backends with **AWS Lambda**, **API Gateway**, **DynamoDB**, secured by **Amazon Cognito**, **AWS** Signature Version 4, and **Cognito Identity Pools**.
- Modularized the **React** frontend using **Webpack 5 module federation**, with collaborative **Git source control** and deployment workflows through **AWS CodeCommit**.
- Built serverless routing with UUIDs, **TypeScript** decorators, and **Express**-style Lambda middlewares.
- Monitored Lambdas using **CloudWatch**, integrated UI alerts via **React Toastify** and **AWS SNS**.
- Implemented dynamic threshold system with **DynamoDB** TTL, conditional writes, and **TypeScript** models.
- Created shipping calculator using **Step Functions**, **DynamoDB**, Lambda Map states, and JSONPath.
- Integrated metadata tooltips via **AWS S3** and Lambda, rendered in **React** with Tooltip.js and Framer Motion.
- Used **Yup**, **Formik**, **Zod** for form validation; ensured backend type safety.
- Managed async workflows via **AWS SQS**, SNS, and FIFO queues.
- Delivered production code through **GitHub Actions** and **AWS Amplify** using structured **Git branching**, **pull requests**, and **peer reviews**.
- Built utility libraries using **Lodash**, Regex, **TypeScript** utility types.
- Synced pricing with **DynamoDB** Streams, Lambda consumers.

- Automated events with **AWS EventBridge**, custom buses, schema registries, and retries.
- Implemented **automated unit testing** and **integration testing** with **Jest**, **React Testing Library**, and **Supertest**, with coverage checks integrated into CI pipelines.

Senior UI Developer for TD Bank

- Built UIs using **React**, **TypeScript**, **JavaScript**, **GraphQL**, **Kafka**, and **Jest** on **Azure**.
- Refactored forms with **Formik**, **Yup**, Regex, **Lodash**.
- Managed state using **Redux Toolkit**, Thunk, and Persist.
- Integrated **graphql-request**, **Apollo Client** with **Azure App Service GraphQL** APIs.
- Improved **user experience (UX)** and frontend performance using **react**-to-print, useMemo, useCallback, and reusable custom Hooks.
- Created reusable dropdowns using data from **Azure Service Bus**.
- Built modular apps on **AKS**, enabled autoscaling and **microservice architecture**.
- Shared components via **Webpack 5 module federation**.
- Worked in an **Agile team environment**, tracking defects through **Bitbucket** Issues and contributing to iterative development and maintenance of complex production applications.
- Designed **GraphQL** schemas and used **Bitbucket Git** repositories for collaborative branching, **pull requests**, and code reviews.
- Built CI pipelines using **Azure DevOps**, **Jest**, and **React Testing Library** to automate application checks and support reliable releases.
- Secured APIs via **Azure API Management**, scaled with **AKS** autoscaling.
- Implemented **workflow automation** using **Azure Logic Apps** and custom **JavaScript** connectors.

Full Stack Delivery

- Delivered enterprise frontend and backend capabilities across **React**/**TypeScript** applications, serverless APIs, data persistence, automated tests, and cloud release pipelines for Air Canada and TD Bank.

Inovi Technologies

Senior Software Developer

09 Sep 2015 - 15 Jan 2020

- Built scalable serverless apps using **AWS Lambda**, **API Gateway**, and **AWS Amplify**, with core services in **TypeScript** for strong typing and maintainability.
- Optimized front-end performance for **Angular 8** and **React** apps hosted on **Amazon S3**, distributed via **CloudFront** with cache invalidation.
- Delivered production full-stack applications using **React**, **Angular**, **TypeScript**, **AWS Lambda**, **API Gateway**, and S3, supporting high availability and cost-efficient application delivery.
- Implemented **microservices** using **AWS Lambda**, **API Gateway**, **Step Functions**, and **Amazon SQS**, following **event-driven architecture**.
- Secured the **React** admin portal using **AWS Cognito** (MFA, federated identities), **Secrets Manager**, and **CloudFormation** for IaC.
- Built **REST APIs** using **API Gateway** + Lambda, with Lambda authorizers, mapping templates, and Stage Variables for routing.
- Reused UI components across **Angular** and **React** applications to improve development efficiency, maintainability, and consistency.
- Managed sessions, access control, and identity federation using **AWS Cognito**, **OIDC**, **OAuth2**, Amplify Auth, and **Cognito Identity Pools**.
- Monitored Lambda performance using **CloudWatch**, custom metrics, alarms, and log insights for proactive debugging.
- Fine-tuned frontend performance using **React Hooks**, useMemo, **NgRx**, and **RxJS** for efficient API interaction.
- Used **DynamoDB** for low-latency **NoSQL** and **Amazon S3** for static content, with fine-grained access via IAM and bucket policies.
- Automated **CI/CD** pipelines with **AWS CodePipeline**, **CodeBuild**, **CodeDeploy**, and **GitHub source control** to support deployments across multiple environments.

- Traced and debugged with **AWS X-Ray**, analyzing Lambda latencies and async execution flows.
- Integrated **AWS SES** for transactional emails, verification, and user notifications with raw/template formats.
- Delivered architectural solutions using Lambda, **API Gateway**, S3, **DynamoDB**, and enforced cost-efficient scalability.
- Used **AWS SAM** and **CloudFormation** for infrastructure provisioning, **CI/CD**, and repeatable, auditable deployments.
- Applied **TypeScript** best practices, **modular software design**, and validation hooks, supported by **automated unit testing** and **integration testing** using **Jest** and **Supertest**.
- Applied event-driven and **serverless design patterns** across frontend applications, APIs, messaging, data storage, authentication, and infrastructure automation to build maintainable full-stack solutions.

Softbrij IT Solutions

Software Developer

07 Sep 2011 - 01 Sep 2015

Backend Developer for Client One Internal Site

- Built scalable **Node.js**/**Express REST APIs** for a high-traffic **AngularJS** frontend on **Azure Web Apps**, handling 35,000+ concurrent users with low latency.
- Developed automation and monitoring tools in **Node.js**, improving QA and uptime tracking; increased CTR by 4% using **Azure Application Insights**.
- Deployed and optimized **REST APIs** via **Azure API Management**, improving reporting speed by 20% with endpoint tuning and caching.
- Modernized a **legacy web application** into a **single-page application** served through **Azure CDN**, reducing load times and improving client-side performance.
- Integrated **Azure Blob Storage** for efficient static asset delivery.
- Developed secure, token-based services using **Azure Active Directory** (AAD) with enterprise-grade **RBAC**.
- Built **CI/CD** pipelines to automate application build, testing, and deployment stages.
- Tuned queries on **Azure SQL Database** using indexing and execution plans for high-load dashboards.
- Integrated application monitoring, logs, alerts, and diagnostics to support proactive issue resolution.

Full Stack Developer for Client Two

- Built responsive **AngularJS** components for a customer-facing app hosted on **Amazon EC2**, accelerated via **Amazon CloudFront**.
- Worked in an **Agile team environment** using **Jira**, contributing to sprint planning, daily standups, and iterative delivery of full-stack application features.
- Created a test coverage tool using **AWS Lambda** and **Node.js**, reducing customer complaints by 5%.
- Integrated **REST APIs** via **Amazon API Gateway**, with backend on EC2 and data in **Amazon RDS** (**PostgreSQL**).
- Delivered mobile-optimized UIs using **AngularJS**, hosted on **Amazon S3**, balanced through **Elastic Load Balancer** (ELB), saving time for 27,000+ users.
- Mentored junior devs on **AngularJS** and **Git** workflows using **AWS CodeCommit**.
- Improved backend scalability and monitoring using **Node.js**/**Express** and **Amazon CloudWatch** for logs and performance alerts.
- Increased fault tolerance with **Amazon ELB** for automatic traffic routing and failover.
- Implemented **CI/CD** pipelines via **AWS Elastic Beanstalk**, enabling automated provisioning and rollbacks.
- Provided real-time operational visibility using **Amazon CloudWatch Logs** for faster incident response and tuning.

Education and Training

Bachelor of Technology in Computer Science and Engineering

All India Institute of Technology | 09 Jun 2007 - 10 May 2011 | EQF Level 6

Mobile Applications and Web Development

TAV College | Montreal, Canada | 02 Feb 2020 - 24 Jan 2022